

HDFC BANK — CUSTOM LLM PIPELINE & RAG PLATFORM

End-to-End Technical Architecture, Evaluation Gate, Model Registry & Production Deployment Report

1. Executive Overview & System Objective

The **HDFC Custom LLM Pipeline** is an enterprise-grade Artificial Intelligence platform built specifically for HDFC Bank. It provides a complete 0-to-end framework for ingesting official banking policy documents (PDF/DOCX/TXT), generating vector embeddings, building context-adapted model adapters, running automated compliance Evaluation Gates, managing model registry cards, deploying canary rollouts, and executing real-time grounded inference with 100% data persistence.

Core Architecture Goal: Deliver zero-hallucination, highly compliant customer banking assistance while maintaining 0 MB local RAM overhead on cloud web services (eliminating container OOM crashes) and enforcing strict regulatory guardrails for fraud escalation, investment disclosure, and legal advice refusal.

2. End-to-End Technology Stack

Layer	Technologies & Frameworks	Architectural Purpose
Frontend UI	HTML5, Vanilla CSS, JS ES6+, Lucide Icons, Canvas Sparklines	Single-page executive dashboard, interactive Playground, dynamic themes, sparkline telemetry
Backend API	Python 3.11, FastAPI, Uvicorn ASGI Server, Pydantic v2	High-performance asynchronous REST API routing, schema validation, thread locking
Database Storage	PostgreSQL (Render Cloud DB) + SQLAlchemy ORM	Permanent cloud storage for datasets, runs, evaluations, registry cards, deployments, and logs
Cloud GPU LLM	Hugging Face InferenceClient (Meta Llama-3.3-70B & Qwen2.5-32B)	High-speed (300ms) serverless GPU inference for real multi-step model text generation
Vector Retriever	PureNumpyEmbedder (L2-normalized TF-IDF) + bge-small-en-v1.5	<25MB RAM vector embedding matrix indexing, sha256 chunk deduplication, cosine similarity
Document Ingestion	PyPDF, python-docx, Regex PII Redaction Engine	Page-aware document parsing, automatic PII redaction (PAN, Aadhaar, Phone Numbers)

3. Detailed Module Breakdown (0-to-End Workflow)

Module A: Document & Dataset Ingestion

- **Page-Aware Chunking:** PDF and DOCX files are parsed with exact page boundary tracking, maintaining page numbers for precise source citations.
- **PII Redaction:** Automatic regex filters detect and redact sensitive customer data (PAN Cards, Aadhaar Numbers, Phone Numbers) prior to vector indexing.
- **SHA256 Deduplication:** Unique chunk content hashes prevent duplicate document uploads from swelling the database.

Module B: Vector Retriever & Grounded RAG

- **Lightweight Vector Indexer:** Computes normalized dense vector matrices (`retriever.pkl`) in 0.1 seconds with less than 25MB RAM overhead.
- **Semantic Top-K Retrieval:** Performs cosine matrix multiplication against user queries, injecting top matching chunks into the model context block.
- **Confidence Scoring:** Calculates a 0-100% confidence heuristic based on vector similarity score and answer word overlap.

Module C: AI Factory & Context Adaptation

- **System Prompt Compiler:** Assembles approved dataset records with `STRICT\_GROUNDING\_INSTRUCTIONS` and default phrasing examples.
- **Zero-Fine-Tuning Architecture:** Implements prompt-based context-adaptation, providing full model customization without expensive GPU retraining.

Module D: Automated Evaluation Gate

- **5 Compliance Fixtures:** Tests fraud escalation, investment refusal, legal advice refusal, and factual grounding.
- **GPU Accelerated Benchmark:** Executes 10 evaluation test calls in 3 seconds total using cloud GPU serverless inference.
- **Deterministic Pass/Fail Gate:** Models must pass 100% of critical compliance test cases to be eligible for production promotion.

Module E: Model Registry & Canary Deployments

- **Governance Model Cards:** Stores Intended Use, Prohibited Use, Gate Pass status, and owner credentials.
- **Canary Rollout Engine:** Supports progressive traffic percentage allocation (10% to 100%) with 1-click instant rollback capability.

4. Enterprise Security, Employee Verification & Password Reset

To enforce enterprise security, the platform integrates a PostgreSQL-backed **Authorized HDFC Employee Directory**. User registration and profile updates validate employee IDs against pre-approved bank credentials, binding all user activity to a verified officer ID.

Division + Numeric Password Reset Flow: Users can recover credentials by providing their Division Code (e.g., `DEV`, `AI`, `EMP`, `GOV`, `SEC`) and Numeric Code (e.g., `3301`, `101`, `4829`, `9901`). The backend normalizes the codes to `HDFC-{DIV}-{NUM}` and verifies authorization in real time.

Authorized HDFC Unique Employee Directory Table

Division Code	Employee ID	Full Name & Role	Department
AI	HDFC-AI-101	Abhi (Lead AI Engineer & System Admin)	AI & Machine Learning
AI	HDFC-AI-102	Senior AI Systems Architect	AI & Machine Learning
AI	HDFC-AI-103	RAG & LLM Alignment Specialist	AI & Machine Learning
EMP	HDFC-EMP-4829	Senior Banking Operations Manager	Banking Operations
EMP	HDFC-EMP-4830	Branch Operations Executive	Banking Operations
EMP	HDFC-EMP-5102	Fixed Deposit & Savings Specialist	Banking Operations
GOV	HDFC-GOV-9901	Chief Risk & Compliance Officer	Governance & Compliance
GOV	HDFC-GOV-9902	Nodal Grievance Redressal Officer	Governance & Compliance
SEC	HDFC-SEC-7701	Senior Fraud Investigation Lead	Cybersecurity & Fraud
SEC	HDFC-SEC-7702	Information Security & Access Lead	Cybersecurity & Fraud
DEV	HDFC-DEV-3301	Core Pipeline Integration Developer	Enterprise IT
DEV	HDFC-DEV-3302	Cloud Platform Operations Engineer	Enterprise IT

5. Live Deployment URLs & Production Verification

- **Live Web Frontend:** <https://jarvisggits47.github.io/HDFC-Custom-LLM-Pipeline/>
- **Live Backend API (Render):** <https://hdfc-custom-llm-backend.onrender.com>
- **PostgreSQL Database:** Render Managed PostgreSQL (`hdfc\_db` / zero data loss across container restarts)
- **GitHub Repository:** <https://github.com/Jarvisggits47/HDFC-Custom-LLM-Pipeline>

PROJECT COMPLETION SUMMARY:

The HDFC Custom LLM Pipeline successfully fulfills all functional, architectural, compliance, and governance requirements. It features zero OOM memory protection, high-speed GPU serverless inference, RAG document grounding with page citations, automated Evaluation Gate scoring, model cards, canary deployment control, and PostgreSQL database employee verification.